

## **Prep Test 2: Answer Key and Rubric**

75 minutes | 100 points | Keep separate from the student test.

Give follow-through credit: deduct an originating error once, not again for consistent later execution. Score missing parts separately. Do not require unrequested explanations or exact polynomial extrema.

### **1. Finish a piecewise graph (10 points)**

1. For  $-2$  use  $2x+4$ :  $p(-2)=0$ . At  $-1$  and  $2$  use the constant branch: both values are  $3$ .
2. Draw  $y=2x+4$  to the left of an open  $(-1,2)$ . Draw  $y=3$  to the right of a closed  $(-1,3)$ . The horizontal branch includes its boundary.
3. Every real  $x$  belongs to a branch, so the domain is all real numbers. The left branch gives every  $y$  below  $2$ ; the right branch adds only  $3$ . Range:  $(-\infty, 2) \cup \{3\}$ .

#### **Scoring**

- 3 points: Three evaluations, 1 each.
- 2 points: Correct branch shapes and sides, 1 each.
- 2 points: Boundary coordinates and open/closed marks, 1 per boundary.
- 3 points: Domain 1; complete range including its gap 2.

### **2. Transform both coordinates of a given graph (12 points)**

1. For source point  $(u,v)$ , the new point is  $(u+2, 1-v)$ : shift right 2, reflect across the  $x$ -axis, then shift up 1.
2. The vertices become  $(-1,0)$ ,  $(1,3)$ ,  $(4,1)$ , and  $(6,-1)$ . Join in order with straight segments and included endpoints.
3. Domain  $[-1,6]$ . The original range  $[-2,2]$  becomes  $[-1,3]$  after reflection and vertical shift.

#### **Scoring**

- 4 points: Four transformed vertices, 1 each.
- 2 points: Correctly connected straight segments. Preserve follow-through credit from an earlier coordinate error.
- 2 points: All coordinate labels 1; both included endpoints 1.
- 2 points: Domain.
- 2 points: Range.

### 3. Graph both composition orders with a reflection (14 points)

1. For source  $(u,v)$ , solve  $1-(3/5)x=u$ , so  $f(g(x))$  maps to  $((5/3)(1-u),v)$ . Left-to-right vertices:  $(-5,2)$ ,  $(-5/3,0)$ ,  $(10/3,3)$ ,  $(20/3,-1)$ . Domain  $[-5,20/3]$ .
2. For  $g(f(x))$ , map  $(u,v)$  to  $(u,1-(3/5)v)$ . Vertices:  $(-3,8/5)$ ,  $(-1,-4/5)$ ,  $(2,1)$ ,  $(4,-1/5)$ . Domain  $[-3,4]$ .
3. Connect each vertex set in its own left-to-right order. Use solid and dashed lines as specified and mark all endpoints included.
4. For  $f(g(x))$ , reflect horizontally, stretch horizontally by  $5/3$ , then shift right  $5/3$ . For  $g(f(x))$ , reflect vertically, compress vertically by  $3/5$ , then shift up 1. Equivalent correctly ordered descriptions are accepted.

#### Scoring

- 2 points: Valid coordinate work for both composition orders, 1 each. Accept mapping rules, substitution equations, or equivalent calculations; no particular method is required.
- 6 points: Four transformed vertices for each composition, 0.75 each.
- 3 points: Correct segment connections and included endpoints 2; all coordinate labels, graph names, and specified line styles 1 combined.
- 2 points: Domains, 1 each.
- 1 points: Correct transformation description for each composition, 0.5 each.

### 4. Simplify and solve a rational expression (10 points)

1. Factor the numerator as  $(x-2)(x+1)$  and denominator as  $(x-2)(x+3)$ . Simplify to  $(x+1)/(x+3)$ , retaining  $x \neq 2$  and  $x \neq -3$ . Domain:  $(-\infty,-3) \cup (-3,2) \cup (2,\infty)$ .
2. Solve  $2(x+1)=x+3$  to get  $x=1$ . It is in the original domain and gives  $R(1)=1/2$ .
3. S excludes only  $-3$ . R also excludes  $2$ , where its original denominator is zero. Thus the domains differ, even though their values agree wherever R is defined.

#### Scoring

- 5 points: Factoring 2 (1 per polynomial); simplification 1; original domain 2 (1 per exclusion). Accept equivalent domain notation.
- 3 points: Valid equation work 2; solution  $x=1$ , 1. Preserve valid algebraic follow-through from an earlier error.
- 2 points: Correct domain comparison 1; explains original denominator exclusion at  $x=2$ , 1. Do not deduct again for the same restriction error from part (a).

## 5. Compare two algebraic compositions (6 points)

1.  $A(B(x)) = (2x-1)^2 - 2(2x-1) = 4x^2 - 8x + 3$ .
2.  $B(A(x)) = 2(x^2 - 2x) - 1 = 2x^2 - 4x - 1$ .
3. Equality requires  $2x^2 - 4x + 4 = 0$ , or  $(x-1)^2 + 1 = 0$ . A real square cannot equal  $-1$ , so there are no real solutions. Equivalently, the quadratic discriminant is negative.
4. The difference  $A(B(x)) - B(A(x)) = 2(x-1)^2 + 2$  is always positive, so the first composition is always greater. This observation is a valid justification, not an extra student requirement.

### Scoring

- 2 points:  $A(B(x))$ : full substitution 1; correct expansion 1.
- 2 points:  $B(A(x))$ : full substitution 1; correct simplification 1.
- 1 points: Correct equality equation and valid real-solution reasoning. Preserve algebraic follow-through from one earlier error.
- 1 points: Explicitly concludes no real solutions.

## 6. Show a flattened crossing in a fifth-degree polynomial (14 points)

1.  $P(x) = a(x+2)^2(x-1)^3$ . From  $P(0) = -8$ ,  $-4a = -8$ , so  $a = 2$ .
2.  $P(x) = 2(x+2)^2(x-1)^3$  has positive leading coefficient and odd degree: left end down, right end up. Touch the axis at  $(-2, 0)$  without changing sign; cross with horizontal flattening at  $(1, 0)$ . Label  $(0, -8)$ .
3. The squared factor is positive except at  $-2$ ; the cubic factor is negative below  $1$  and positive above  $1$ .  $P > 0$  on  $(1, \infty)$ .  $P < 0$  on  $(-\infty, -2)$  union  $(-2, 1)$ . Exclude both zeros from strict sign intervals.

### Scoring

- 4 points: Multiplicity factors 2; leading-factor calculation and final equation 2.
- 1 points: Correct labeled y-intercept.
- 3 points: Both labeled zeros 1; touch at  $-2$  1; clearly flattened crossing at  $1$  1.
- 2 points: Both end directions, 1 each.
- 4 points: Positive interval 1; negative intervals 2; explanation of sign change/no change 1.

## 7. Keep all restrictions in a composed quotient (8 points)

1.  $H(x) = \sqrt{9-x^2} / (4-x^2)$ .
2. The square root requires  $9-x^2 \geq 0$ , giving  $-3 \leq x \leq 3$ . The denominator requires  $4-x^2 \neq 0$ , so exclude  $-2$  and  $2$ .
3. Domain  $[-3, -2)$  union  $(-2, 2)$  union  $(2, 3]$ . At  $3$ ,  $f(g(3)) = \sqrt{0} = 0$  and the denominator is  $-5$ , so it is allowed. At  $2$ ,  $g(2) = 5$ , so the original denominator is zero and it is not allowed.

### Scoring

- 2 points: Correct substitution in numerator 1 and denominator 1.
- 2 points: Radical inequality and bounded restriction.
- 1 points: Both denominator exclusions, 0.5 each.
- 2 points: Final interval intersection including the allowed endpoints.
- 1 points: Boundary checks and explanations, 0.5 each.

## 8. Sketch a quartic with crossings and a touch (8 points)

1. Degree 4, leading coefficient -1. Zeros -2 and 3 are simple; 1 has multiplicity 2.  $R(0) = -(2)(1)(-3) = 6$ , so label (0,6).
2. Both ends fall. Cross upward at (-2,0), stay above the axis, touch and turn at (1,0), stay above until crossing downward at (3,0). Draw a smooth graph with a peak on each side of the touch; exact peaks are not requested.
3.  $R > 0$  on  $(-2, 1) \cup (1, 3)$ ;  $R < 0$  on  $(-\infty, -2) \cup (3, \infty)$ . The even root does not change sign but is excluded from the strict-positive intervals.
4. Even degree makes the signs at the two far ends the same; the negative leading coefficient makes both ends fall. Odd degree makes the two ends opposite.

### Scoring

- 1 points: Degree/leading coefficient 0.5; zeros and multiplicities 0.5.
- 1 points: Calculated and labeled y-intercept.
- 2 points: Labeled zeros 0.5; crossing/touch behavior 1; correctly signed smooth connections 0.5.
- 1 points: Both ends down.
- 2 points: Complete positive and negative interval lists, 1 each.
- 1 points: Even versus odd degree end-behavior explanation.

## 9. Use a composite model to meet a budget (10 points)

1. Use  $C(A(w))$ : A converts width in feet to area in square feet; C converts area to dollars.
2.  $A(w) = 12w - w^2 = 36 - (w-6)^2$ . On  $[2, 6]$ , area increases from 20 to 36.  $A = 30$  gives  $(w-6)^2 = 6$ , so the relevant breakpoint is  $b = 6 - \sqrt{6}$ , approximately 3.55.
3. Cost is  $96w - 8w^2$  for  $2 \leq w \leq 6 - \sqrt{6}$ , and  $60w - 5w^2 + 90$  for  $6 - \sqrt{6} < w \leq 6$ . Domain  $[2, 6]$ . Either formula gives \$240 at the breakpoint; respect the original price condition when assigning the boundary.
4. At  $w = 4$ , area is 32, so use the second price branch:  $C(32) = 240 + 5(2) = \$250$ .
5. \$260 is above the \$240 breakpoint price, so the maximum uses the second branch:  $5A + 90 = 260$  gives  $A = 34$ . Then  $36 - (w-6)^2 = 34$  gives  $w = 6 \pm \sqrt{2}$ . Only  $6 - \sqrt{2}$  is in  $[2, 6]$ .
6. The greatest width is  $6 - \sqrt{2}$  feet, approximately 4.59 feet. Area and both price branches are increasing over the permitted widths, so wider permitted beds exceed the budget. Feasible widths are  $[2, 6 - \sqrt{2}]$ .

### Scoring

- 2 points: Correct composition 1; correct unit/order explanation 1.
- 3 points: Width breakpoint 1; two cost expressions 0.5 each; branch intervals and model domain 1.
- 2 points: Area 32 and correct price branch 1; cost \$250 1.
- 3 points: Budget equation/area 0.5; exact width 1; rounded width 0.5; branch/domain check 0.5; explains why it is the greatest width 0.5. Preserve follow-through from one earlier model error.

## 10. Construct and graph a square-root function (8 points)

1. The domain gives  $h=1$ . The upper range boundary gives  $k=3$ , and the function is reflected downward. Substitute  $(5,-1)$ :  $-1=2a+3$ , so  $a=-2$ . Rule  $r(x)=-2\sqrt{x-1}+3$ .
2. Starting point  $(1,3)$ . Three useful additional points are  $(2,1)$ ,  $(5,-1)$ ,  $(10,-3)$ . Draw a decreasing square-root curve, with included start and a rightward continuation.
3. Solve  $-2\sqrt{x-1}+3=0$ :  $\sqrt{x-1}=3/2$ ,  $x-1=9/4$ ,  $x=13/4$ . The x-intercept is  $(13/4,0)$ . The function decreases on  $(1,\infty)$ ; domain-based endpoint-inclusive wording is also acceptable.

### Scoring

- 3 points: Correct  $h$  and  $k$  1; solves for  $a$  1; final rule 1.
- 3 points: Included labeled starting point 1; three additional labeled points 1; correct connected curve and continuation 1.
- 2 points: Exact x-intercept 1; decreasing interval 1. Accept  $(1,\infty)$  or an explicitly stated decrease over its full domain  $[1,\infty)$ .